

CS228 : Logic for Computer Science

Module 1: Propositional Logic

Lecture 6: Completeness of Natural Deduction

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Recap: The Story so far

Module 1 Propositional Logic (Lectures 1–5)

- ▶ L1: Syntax
- ▶ L2: Parse Trees, Semantics & Problems of Interest
- ▶ L3: Formal Proofs & Natural Deduction
- ▶ L4: Natural Deduction Rules
- ▶ L5: Soundness & Completeness

Soundness: Anything that is provable is (semantically) correct

If $F_1, \dots, F_n \vdash G$ is valid, then $F_1, \dots, F_n \models G$ holds

Completeness: Anything that is correct is provable in ND!

If $F_1, \dots, F_n \models G$ holds, then $F_1, \dots, F_n \vdash G$ is valid

Recap: Completeness of Natural Deduction for propositional logic

Theorem (Anything that is correct is provable)

If $F_1, \dots, F_n \models G$ holds, then $F_1, \dots, F_n \vdash G$ is valid

Whenever $F_1, \dots, F_n$ semantically entail G , then G can be derived from $F_1, \dots, F_n$ using the ND rules.

Proof: Let $F_1, \dots, F_n \models G$. Proof is in three steps

- ▶ Step 1: Show that $\models F_1 \rightarrow (F_2 \rightarrow (\dots (F_n \rightarrow G) \dots))$ holds. **This is easy!**
 - ▶ By contradiction, suppose $\not\models F_1 \rightarrow (F_2 \rightarrow (\dots (F_n \rightarrow G) \dots))$.
 - ▶ This nested implication is false only if there exists an assignment s.t. G evaluates to 0 and $F_1, \dots, F_n$ evaluate to 1.
 - ▶ Which contradicts $F_1, \dots, F_n \models G$.
- ▶ Step 2: Show that $\vdash F_1 \rightarrow (F_2 \rightarrow (\dots (F_n \rightarrow G) \dots))$ is valid
- ▶ Step 3: Show that $F_1, \dots, F_n \vdash G$ is valid

Recap: Completeness of Natural Deduction for propositional logic

Theorem (Anything that is correct is provable)

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- ▶ Step 1: Show that $\models F_1 \rightarrow (F_2 \rightarrow (\dots (F_n \rightarrow G) \dots))$ holds.
- ▶ Step 2: Show that $\vdash F_1 \rightarrow (F_2 \rightarrow (\dots (F_n \rightarrow G) \dots))$ is valid
- ▶ Step 3: Show that $F_1, \dots, F_n \vdash G$ is valid **This is also easy!**
 - ▶ Take the proof from Step 2.
 - ▶ Add $F_1, \dots, F_n$ as premises as done in soundness proof.
 - ▶ Then apply *MP* n times to arrive at G .

Recap: Completeness of Natural Deduction for propositional logic

Theorem (Anything that is correct is provable)

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- ▶ Step 1: Show that $\models F_1 \rightarrow (F_2 \rightarrow (\dots (F_n \rightarrow G) \dots))$ holds.
- ▶ Step 2: Show that $\vdash F_1 \rightarrow (F_2 \rightarrow (\dots (F_n \rightarrow G) \dots))$ is valid
- ▶ Step 3: Show that $F_1, \dots, F_n \vdash G$ is valid

So, the only thing left is Step 2. This is the hard part!

Completeness of Natural Deduction for propositional logic

Step 2 follows from the following theorem:

Theorem

For any propositional formula G , if $\models G$ holds, then $\vdash G$ is valid. That is, if G is a tautology, then it is a theorem!

Proof

- ▶ Let $p_1, \dots, p_m$ be the propositional variables in G ; all 2^m valuations make G true.
- ▶ From just this, can derive a proof of G ?
 - ▶ Step 2.1 **Key Idea**: Encode each line of truth table as a sequent!
 - ▶ If in line ℓ of truth table, p_i has value 1, then set $\hat{p}_i = p_i$ otherwise set $\hat{p}_i = \neg p_i$
 - ▶ Sequent for line ℓ : $\hat{p}_1 \dots \hat{p}_m \vdash G$
 - ▶ Step 2.2: Construct proofs for each of 2^m sequents.
 - ▶ For each line ℓ show that the sequent $\hat{p}_1 \dots \hat{p}_m \vdash G$ is valid.
 - ▶ Step 2.3: Assemble them to obtain proof of G

Completeness Proof contd

Lemma

Let F be any propositional formula with variables $p_1, \dots, p_m$. Let ℓ be a line number in Truth Table of F . Let $\hat{p}_i = p_i$ if p_i has value 1 in line ℓ , and $= \neg p_i$ if p_i has value 0 in line ℓ . Then

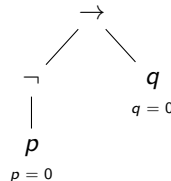
1. $\hat{p}_1, \dots, \hat{p}_m \vdash F$ is valid if F evaluates to 1 in line ℓ
2. $\hat{p}_1, \dots, \hat{p}_m \vdash \neg F$ is valid if F evaluates to 0 in line ℓ

Proof by **Structural Induction** on structure of F , i.e., induction on height of parse tree for F .

An Example

Let $F = \neg p \rightarrow q$, i.e., $F_1 = \neg p$ and $F_2 = q$. We derive the proof for line ℓ below.

p	q	$\neg p$	F	
0	0	1	0	$\leftarrow \ell$
0	1	1	1	
1	0	0	1	
1	1	0	1	



Goal: $\hat{p}, \hat{q} \vdash \neg F$.

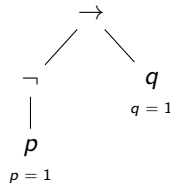
- **Leaves:** $p = 0 \Rightarrow \hat{p} = \neg p$, so $\neg p \vdash \neg p$ (base case).
 $q = 0 \Rightarrow \hat{q} = \neg q$, so $\neg q \vdash \neg q$ (base case).
- **$\neg$ node ($F_1 = \neg p$):** F_1 evaluates to 1 $\Rightarrow$ using the leaf, $\neg p \vdash \neg p$, i.e., $\neg p \vdash F_1$.
- **$\rightarrow$ node (F , root):** F evaluates to 0 at this line. We have $\neg p$ (from the $\neg$ node) and $\neg q$ (from the leaf) as premises — using these, we derive $\neg F$ on the right:

- $\neg p$ premise
- $\neg q$ premise
- $\neg p \rightarrow q$ assumption
- q MP 3,1
- $\perp$ $\neg e$ 4,2
- $\neg(\neg p \rightarrow q)$ $\neg i$ 3-5

An Example (contd.)

Same $F = \neg p \rightarrow q$ ($F_1 = \neg p$, $F_2 = q$). Now let's try line ℓ : $p = 1, q = 1$.

p	q	$\neg p$	F	
0	0	1	0	
0	1	1	1	
1	0	0	1	
1	1	0	1	$\leftarrow \ell$



Goal: $\hat{p}, \hat{q} \vdash F$.

- ▶ **Leaves:** $p = 1 \Rightarrow \hat{p} = p$, so $p \vdash p$ (base case);
 $q = 1 \Rightarrow \hat{q} = q$, so $q \vdash q$ (base case) — both leaves now positive.
- ▶ **$\neg$ node ($F_1 = \neg p$):** F_1 now evaluates to 0 (since $p = 1$)! By IH, $p \vdash p$; use $\neg\neg i$ to get $p \vdash \neg\neg p$, i.e., $p \vdash \neg F_1$.
- ▶ **$\rightarrow$ node (F , root):** F_1 evaluates to 0, F_2 evaluates to 1 — a third sub-subcase! We have $p \vdash \neg F_1$ and $q \vdash F_2$ as premises — using these, we derive F on the right:

1.	p	premise
2.	q	premise
3.	$\neg\neg p$	$\neg\neg i$ 1
4.	$\neg p$	assumption
5.	$\perp$	$\neg e$ 4,3
6.	q	$\perp e$ 5
7.	$\neg p \rightarrow q$	$\rightarrow i$ 4-6

Completeness Proof contd

Lemma

Let F be any propositional formula with variables $p_1, \dots, p_m$. Let ℓ be a line number in Truth Table of F . Let $\hat{p}_i = p_i$ if p_i has value 1 in line ℓ , and $= \neg p_i$ if p_i has value 0 in line ℓ . Then

1. $\hat{p}_1, \dots, \hat{p}_m \vdash F$ is valid if F evaluates to 1 in line ℓ
2. $\hat{p}_1, \dots, \hat{p}_m \vdash \neg F$ is valid if F evaluates to 0 in line ℓ

Proof by **Structural Induction** on structure of F , i.e., induction on height of parse tree for F .

- ▶ Base Case : If $F = p$, a proposition, then $p \vdash p$ and $\neg p \vdash \neg p$.
- ▶ Induction Hyp: Assume Lemma for formulae with parse tree height $\leq k - 1$
- ▶ Case Negation : $F = \neg F_1$
 - ▶ If F evaluates 1 at line ℓ , then F_1 evaluates to 0. By **inductive hypothesis**, $\hat{p}_1, \dots, \hat{p}_m \vdash \neg F_1$, i.e., $\hat{p}_1, \dots, \hat{p}_m \vdash F$.
 - ▶ If F evaluates 0 at line ℓ , then F_1 evaluates to 1. By **inductive hypothesis**, $\hat{p}_1, \dots, \hat{p}_m \vdash F_1$. Use $\neg\neg i$ rule to obtain proof of $\hat{p}_1, \dots, \hat{p}_m \vdash \neg\neg F_1$, i.e., $\hat{p}_1, \dots, \hat{p}_m \vdash \neg F$.

Completeness Proof contd

Case $\rightarrow$: $F = F_1 \rightarrow F_2$

Two subcases:

- ▶ Subcase 1: F evaluates to 0 in line ℓ , (exactly our first example!)
 - ▶ then F_1 evaluates to 1 and F_2 to 0.
 - ▶ Let $\{q_1, \dots, q_k\}$ be the variables of F_1 and let $\{r_1, \dots, r_j\}$ be the variables in F_2 ; $\{q_1, \dots, q_k\} \cup \{r_1, \dots, r_j\} = \{p_1, \dots, p_m\}$.
 - ▶ By inductive hypothesis, $\hat{q}_1, \dots, \hat{q}_k \vdash F_1$ and $\hat{r}_1, \dots, \hat{r}_j \vdash \neg F_2$.
 - ▶ Then, $\hat{p}_1, \dots, \hat{p}_m \vdash F_1 \wedge \neg F_2$.
 - ▶ **Exercise 6.1:** Prove that $F_1 \wedge \neg F_2 \vdash \neg(F_1 \rightarrow F_2)$ and conclude this case!
- ▶ Subcase 2: F evaluates to 1 in line ℓ , there are 3 subsubcases.
 - ▶ If both F_1, F_2 evaluate to 1, then $\hat{p}_1, \dots, \hat{p}_m \vdash F_1 \wedge F_2$.
Exercise 6.2(i): Prove $F_1 \wedge F_2 \vdash F_1 \rightarrow F_2$ and conclude.
 - ▶ If both F_1, F_2 evaluate to 0, then $\hat{p}_1, \dots, \hat{p}_m \vdash \neg F_1 \wedge \neg F_2$.
Exercise 6.2(ii): Prove $\neg F_1 \wedge \neg F_2 \vdash F_1 \rightarrow F_2$ and conclude.
 - ▶ If F_1 evaluates to 0 and F_2 evaluates to 1, (our second example!) then $\hat{p}_1, \dots, \hat{p}_m \vdash \neg F_1 \wedge F_2$.
Exercise 6.2(iii): Prove $\neg F_1 \wedge F_2 \vdash F_1 \rightarrow F_2$, and complete this case.

Completeness proof (Contd.)

Exercise 6.3

- ▶ Complete similar proofs for the cases $\wedge, \vee$.

Or at least read it from Huth & Ryan's book!

This completes the proof of the Lemma, finishing Step 2.2 of the overall proof.

Completeness of Natural Deduction for propositional logic

Back to Step 2: If $\models G$ holds, then $\vdash G$ is valid; if G is a tautology, then it is a theorem!

Proof

- ▶ Let $p_1, \dots, p_m$ be the propositional variables in G ; all 2^m valuations make G true.
- ▶ From this, we derive a proof of G
 - ▶ Step 2.1 **Key Idea**: Encode each line of truth table as a sequent!
 - ▶ If in line ℓ of truth table, p_i has value 1, then set $\hat{p}_i = p_i$ otherwise set $\hat{p}_i = \neg p_i$
 - ▶ Sequent for line ℓ : $\hat{p}_1 \dots \hat{p}_m \vdash G$
 - ▶ Step 2.2: Use Previous Lemma to Construct proofs for each of 2^m sequents.
 - ▶ For each line ℓ , by Previous Lemma, we have $\hat{p}_1 \dots \hat{p}_m \vdash G$ is valid.
 - ▶ Step 2.3: Assemble the sequents to obtain proof of G
 - ▶ Use **LEM** repeatedly on all the propositional variables of G and $\vee$ -e to obtain $\vdash G$.

An example

Consider the tautology $F = p \wedge q \rightarrow p$.

Exercise 6.4 Show that it is a theorem by applying all steps of proof above.

- ▶ Has two propositions p, q . So 4 valuations
- ▶ Then applying Lemma we have (derive them!):
 - ▶ $p, q \vdash p \wedge q \rightarrow p$
 - ▶ $\neg p, q \vdash p \wedge q \rightarrow p$
 - ▶ $p, \neg q \vdash p \wedge q \rightarrow p$
 - ▶ $\neg p, \neg q \vdash p \wedge q \rightarrow p$
- ▶ Using LEM to show $\vdash F$...

An example: Assembling the Proof

Assemble the four sequents via LEM and $\forall e$, applied three times — exactly as in Huth & Ryan:

1. $p \vee \neg p$ LEM
2. p assumption
3. $q \vee \neg q$ LEM
4. q assumption
5. $p \wedge q \rightarrow p$ Use Lemma
6. $\neg q$ assumption
7. $p \wedge q \rightarrow p$ Use Lemma
8. $p \wedge q \rightarrow p$ $\forall e$ 3,4-5,6-7
9. $\neg p$ assumption
10. $q \vee \neg q$ LEM
11. q assumption
12. $p \wedge q \rightarrow p$ Use Lemma
13. $\neg q$ assumption
14. $p \wedge q \rightarrow p$ Use Lemma
15. $p \wedge q \rightarrow p$ $\forall e$ 10,11-12,13-14
16. $p \wedge q \rightarrow p$ $\forall e$ 1,2-8,9-15

Conclusion!

Propositional Logic is **sound and complete**.

This means we can freely switch between semantic ($\models$) and syntactic ($\vdash$) reasoning — pick whichever is easier for the problem at hand!

Are there other proof systems

- ▶ What we have looked at is called Gentzen-style Natural Deduction.
- ▶ Was introduced in 1934!

Hilbert-style Natural Deduction Proof System

- ▶ Has only 3 axioms and ONE inference rule! Modus Ponens
- ▶ Everything is derived from this...
- ▶ As a result derivations are long and hard, but showing structural induction is easy!

There are MANY more proof systems....

Intuitionistic Logic: No LEM! No double-negation elimination!

- ▶ If you want to prove F , you should prove F , not disprove $\neg F$!
- ▶ No proof by contradiction also!

Exercise 6.5 Why are Intuitionistic logics interesting? What was Brouwer-Hilbert controversy?

A notable point

Showing invalidity of a sequent

- ▶ To show a sequent $F_1, \dots, F_n \vdash G$ is valid, you just derive a proof.
- ▶ How do you show that it is NOT valid?

Use soundness!

1. Just find an assignment such that $F_1 = 1, \dots, F_n = 1$ and $G = 0$.
2. This means $F_1 \dots F_n \not\models G$.
3. Now soundness implies: if $F_1 \dots F_n \vdash G$ is valid, then $F_1 \dots F_n \models G$ holds; the contrapositive gives: if $F_1 \dots F_n \not\models G$, then $F_1 \dots F_n \not\vdash G$.
4. This completes the proof $F_1 \dots F_n \not\vdash G$.

Exercise 6.6: Prove or disprove: $p \vdash p \wedge q$ valid.

Another note regarding reusing what has been derived: The Copy Rule

Consider the theorem

$$\vdash p \rightarrow (q \rightarrow p)$$

1.	p	assumption
2.	q	assumption
3.	p	copy 1
4.	$q \rightarrow p$	$\rightarrow i$ 2-3
5.	$p \rightarrow (q \rightarrow p)$	$\rightarrow i$ 1-4

So, when can you copy? From above ... unless they depend on temporary assumptions whose box has been closed!